Algebra 1
Unit 1
Lesson 1 Practice

Name______________________
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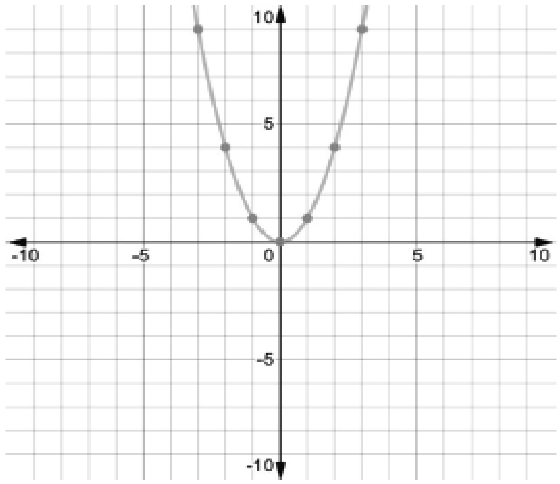
1. A graph is shown.



The graph is the parent function for a(n)

- linear function.
- quadratic function.
- cubic function.
- square root function.
- cube root function.
- absolute value function.
- exponential function.

2. The equation $y = \sqrt{x}$ is the parent function for a(n)

- linear function.
- quadratic function.
- cubic function.
- square root function.
- cube root function.
- absolute value function.
- exponential function.

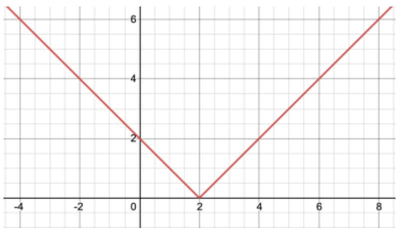
3. The equation $y = |x|$ is the parent function for a(n)

- linear function.
- quadratic function.
- cubic function.
- square root function.
- cube root function.
- absolute value function.
- exponential function.

For equations 4-10, classify the function family for each equation or graph.

4. $y = \frac{3}{5}x - 9$ Linear

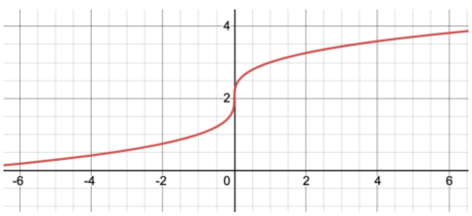
5.



Absolute Value

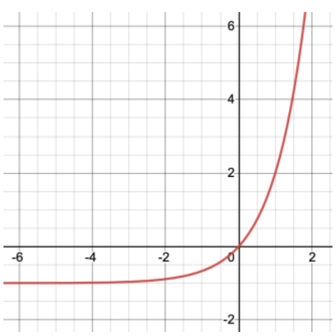
6. $y = -3\left(\frac{1}{2}\right)^x$ Exponential

7.



Cube root

8.



Exponential

9. $y = (x - 3)^3 + 2$ Cubic

10. $y = -4\sqrt{x} + 1$ Square root

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Lesson 2 Practice Period__________________

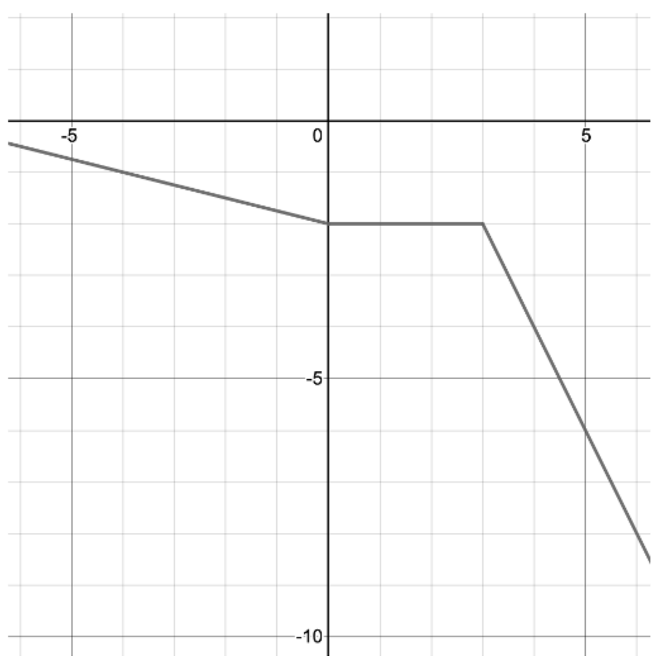
Given the function $r(x) = 5x - 3.5$, find each value.

1. $r(3) = 5(3) - 3.5$
$= 15 - 3.5$
$= 11.5$

2. $r\left(6\frac{1}{2}\right) = 5\left(6\frac{1}{2}\right) - 3.5$
$= 32.5 - 3.5$
$= 29$

3. For what value of x is $r(x) = -21$?
$-21 = 5x - 3.5$
$\underline{+3.5 \qquad + 3.5}$
$-\frac{17.5}{5} = \frac{5x}{5}$
$x = -3.5$

Use the graph of $y = f(x)$ to find each value.



4. $f(-4) = -1$

5. $f(1) = -2$

6. For what value of x is $f(x) = -4$?
$x = 4$

Use the table of values representing $m(x)$ to find each value.

$\boldsymbol{x}$	$\boldsymbol{m(x)}$
7	2
-4	-9
0	8
-15	9
4	-5
6.5	0

7. $m(-4) = -9$

8. $m(4) = -5$

9. What is the y-intercept of the function in the table?
(0,8)

10. Given $d(x) = 3x^3 - 4x^2 + 2$, evaluate $d(x)$ for $x = -1$.

$$= 3(-1)^3 - 4(-1)^2 + 2$$
$$= -3 - 4 + 2$$
$$= -7 + 2$$
$$= -5$$

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The function $G(m) = 49.5m + 89.75$ represents the cost of joining the gym in addition to the one time membership fee. The cost, G, is measured in dollars for m months.

1. Find the value of $G(6)$.
   $49.5(6) + 89.75$
   386.75

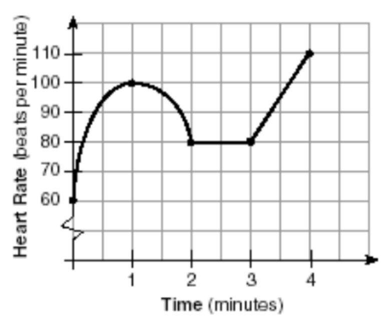
2. What does $G(6)$ represent in the context of the problem?
   6 months of gym membership costs $386.75

3. Is the value of $G(9.75)$ reasonable to interpret in the context of the problem?
   No, customers pay for full months of membership, not partial.

A function $H(m)$ is modeled in the graph below. It shows the heart rate of a swimmer during a 4 minute race where m is the number of minutes and H is the heart rate of the swimmer.

4. Find the approximate value of the graph for $H(1.5)$.

   $H(1.5) \approx 96$



5. What does $H(1.5)$ mean in the context of the problem?
   After 1.5 minutes, the heart rate is 96 BPM.

6. After how many seconds is the swimmers heart rate 80 beats per minute?
   ≈ 10 sec

7. Why does it make sense that $H(2) > H(0)$ in this context?
   After 2 minutes of swimming, the heart rate is higher than before they started swimming.

The table below represents the temperature of a cup of coffee as it begins to cool. Temperature readings are taken every 5 minutes and measured in degrees Fahrenheit (°F).

Minutes, m	0	5	10	15
Temperature, $T(m)$	197	186	170	164

8. Explain the meaning of $T(0) = 197$.
   The initial temperature of the coffee is $197°$.

9. Determine the value of $T(10)$.
   $170°$

10. Interpret $T(10)$ for the given context.
    After 10 minutes of cooling, the temperature of the coffee is $170°$.

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Lesson 4 Practice | Period__________________

For questions 1-5, use the exponent laws to write equivalent expressions.

1. $\left(\frac{6}{5}\right)^{-2} = \frac{6^{-2}}{5^{-2}} \rightarrow \frac{5^2}{6^2}$

2. $\frac{7^7}{7^4} = 7^{7-4} = 7^3$

3. $(5.2 \cdot (-9))^3 = 5.2^3 \cdot (-9)^3$

4. $(-2) \cdot (-2)^{-5} \cdot (-2)^8 = (-2)^{1-5+8} = (-2)^4$

5. $((-4)^4)^9 = (-4)^{4 \cdot 9} = (-4)^{36}$

Find the value of each for questions 6-7.

6. $\sqrt[3]{125} = 125$ $\qquad$ $\sqrt[3]{5 \cdot 5 \cdot 5} = 5$

5 $\quad$ 25

5 $\quad$ 5

7. $\sqrt{25} = 25$ $\qquad$ $\sqrt{5 \cdot 5} = 5$

5 $\quad$ 5

8. Explain why both the $\sqrt{25}$ and $\sqrt[3]{125}$ both equal 5.
Because the square root of 25 is 5, $5 \cdot 5 = 25$ and the cubed root of 125 is 5, $5 \cdot 5 \cdot 5 = 125$. Two of the same factor is the square root and three of the same factor is the cubed root.

For questions 9-10, find the value of the variable that makes the equation true.

9. $\sqrt{c} \cdot \sqrt{c} = 11$ $\sqrt{11} \cdot \sqrt{11} = \sqrt{121} = 11$

$c =$ __11__

10. $\frac{2^3}{3^3} = \left(\frac{3}{2}\right)^g$ $\frac{3^{-3}}{2^{-3}} = \frac{2^3}{3^3}$

$g =$ __-3__

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Lesson 5 Practice | Period__________________

1. In radical form, the expression $x^{\frac{1}{3}}$ is equivalent to __$\sqrt[3]{x}$______.

2. In exponential form, the expression $\sqrt{x}$ is equivalent to __$x^{\frac{1}{2}}$______.

For questions 3-5, rewrite each exponential expression as a radical expression. Then, evaluate the expression.

3. $(-729)^{\frac{1}{3}} = \sqrt[3]{-729}$
$= -9$

-729
$-9 \cdot -81$
$-9 \cdot -9 \quad = -9$

4. $(243)^{\frac{1}{5}} = \sqrt[5]{243}$
$= 3$

243
$9 \cdot 81$
$3 \quad 3$
$9 \cdot 9$
$3 \cdot 3 \quad 3 \cdot 3$

$= \sqrt[5]{3 \cdot 3 \cdot 3 \cdot 3 \cdot 3} = 3$

5. $(225)^{\frac{1}{2}} = \sqrt{225}$
$= 15$

225
$5 \cdot 45$
$5 \cdot 9$
$3 \cdot 3$

$= \sqrt{3 \cdot 3 \cdot 5 \cdot 5}$
$3 \cdot 5 = 15$

6. Simplify the expression using the laws of exponents and then find the value.

$(-27)^{-\frac{2}{9}} \cdot (-27)^{\frac{5}{9}} = (-27)^{-\frac{2}{9}+\frac{5}{9}}$

$(-27)^{\frac{3}{9}}$

$(-27)^{\frac{3\div 3}{9\div 3}}$

$(-27)^{\frac{1}{3}}$

$\sqrt[3]{-27}$

$\sqrt[3]{-3 \cdot -3 \cdot -3}$

$= -3$

The expression $\frac{(-125)^{\frac{11}{9}}}{(-125)^{\frac{8}{9}}}$ can be rewritten using the law of exponents.

7. Which law of exponents can be used to rewrite the expression?
Quotient of Powers law

8. Rewrite the expression as an equivalent expression.
$(-125)^{\frac{11}{9}-\frac{8}{9}}$
$(-125)^{\frac{3}{9}}$
$= (-125)^{\frac{1}{3}}$

9. Write the simplified expression in radical form.
$\sqrt[3]{-125}$

10. Evaluate the radical expression.

125
5 · 25
5 5

$-5 \cdot -5 \cdot -5 = 125$
$\sqrt[3]{-125}$
$= \sqrt[3]{-5 \cdot -5 \cdot -5}$
$= -5$

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1. Explain mathematically why the expressions $\left(a^{\frac{1}{c}}\right)^b$ is equivalent to $(a^b)^{\frac{1}{c}}$ using the expressions $\left(4^{\frac{1}{2}}\right)^5$ and $(4^5)^{\frac{1}{2}}$.

$\left(4^{\frac{1}{2}}\right)^5 = (4)^{5\cdot\frac{1}{2}} = (4)^{\frac{1}{2}\cdot 5} = (4^5)^{\frac{1}{2}}$

2. Identify all of the expressions equivalent to $\left(49^{\frac{1}{2}}\right)^5$.

- $49^{\frac{2}{5}}$
- $(49^5)^{\frac{1}{2}}$
- $\sqrt[5]{49^2}$
- $49^{\frac{5}{2}}$
- $\left(\sqrt{49}\right)^5$

$\left(49^{\frac{1}{2}}\right)^5 = 49^{\frac{5}{2}}$
$= \sqrt[2]{49^5}$
$= \sqrt{49^5}$

For questions 3-7, complete each row with equivalent expressions.

	$a^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^b$	$(a^b)^{\frac{1}{c}}$	$(\sqrt[c]{a})^b$	$\sqrt[c]{a^b}$
3.	$256^{\frac{3}{4}}$	$\left(256^{\frac{1}{4}}\right)^3$	$(256^3)^{\frac{1}{4}}$	$\left(\sqrt[4]{256}\right)^3$	$\sqrt[4]{256^3}$
4.	$64^{\frac{3}{2}}$	$\left(64^{\frac{1}{2}}\right)^3$	$(64^3)^{\frac{1}{2}}$	$\left(\sqrt{64}\right)^3$	$\sqrt{64^3}$
5.	$128^{\frac{5}{7}}$	$\left(128^{\frac{1}{7}}\right)^5$	$(128^5)^{\frac{1}{7}}$	$\left(\sqrt[7]{128}\right)^5$	$\sqrt[7]{128^5}$
6.	$(-27)^{\frac{2}{3}}$	$\left(-27^{\frac{1}{3}}\right)^2$	$(-27^2)^{\frac{1}{3}}$	$\left(\sqrt[3]{-27}\right)^2$	$\sqrt[3]{-27^2}$
7.	$81^{\frac{3}{2}}$	$\left(81^{\frac{1}{2}}\right)^3$	$(81^3)^{\frac{1}{2}}$	$\left(\sqrt{81}\right)^3$	$\sqrt{81^3}$

For questions 8-10, rewrite each expression using a rational exponent in the form $(a)^{\frac{b}{c}}$.

8. $\sqrt[3]{(-44)^5}$

$a = (-44)$

$b = 5$ $\quad = (-44)^{\frac{5}{3}}$

$c = 3$

9. $\left(\sqrt{289}\right)^3$

$a = 289$

$b = 3$ $\quad 289^{\frac{3}{2}}$

$c = 2$

10. $\frac{1}{\sqrt[5]{30^2}}$

$a = 30$

$b = 2$ $\quad 30^{\frac{2}{5}} \rightarrow \frac{1}{30^{\frac{2}{5}}}$ or $30^{-\frac{2}{5}}$

$c = 5$

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1. Write the expression $27^{\frac{2}{3}}$ in two different equivalent forms.

Expression 1	Expression 2
$\sqrt[3]{27^2}$	$\left(27^{\frac{1}{3}}\right)^2$

2. The value of $27^{\frac{2}{3}}$ is

- o 9
- o 32
- o 54
- o 72

For questions 3-5, write an equivalent radical expression and then find the value of the expression.

3. $(-125)^{\frac{2}{3}} = \sqrt[3]{-125^2} = 25$

4. $-\left(\frac{3{,}125}{32}\right)^{\frac{2}{5}} = \sqrt[5]{-\left(\frac{3125}{32}\right)^2} = -\frac{25}{4}$

5. $(81)^{-\frac{3}{2}} = \frac{1}{\sqrt{81^3}} = \frac{1}{729}$

Evaluate each expression in questions 6-10 without a calculator.

6. $(4^{-2})\left(49^{-\frac{1}{2}}\right) = \frac{1}{4^2} \cdot 49^{-\frac{1}{2}} = \frac{1}{16} \cdot \frac{1}{\sqrt{49}} = \frac{1}{16} \cdot \frac{1}{7} = \frac{1}{112}$

7. $\left(256^{\frac{1}{4}}\right)^3 \cdot 6^{-1} = 256^{\frac{3}{4}} \cdot \frac{1}{6} = \sqrt[4]{256^3} \cdot \frac{1}{6} = \sqrt[4]{(4 \cdot 4 \cdot 4 \cdot 4)^3} \cdot \frac{1}{6} = 4^3 \cdot \frac{1}{6} = 64 \cdot \frac{1}{6} = \frac{64}{6} = 10\frac{4}{6} = 10\frac{2}{3} \text{ or } \frac{32}{3}$

8. $-\left(\frac{1}{4}\right)^{-\frac{7}{2}} = \frac{-1^{\frac{-7}{2}}}{4^{\frac{-7}{2}}} = -4^{\frac{7}{2}} = -\sqrt{4^7} = -2^7 = -128$

9. $(-1^{15}) \cdot \left(-64^{\frac{1}{3}}\right) = -1 \cdot \sqrt[3]{-64^1} = -1 \cdot \sqrt[3]{-4 \cdot -4 \cdot -4} = -1 \cdot -4 = 4$

10. $\left(\frac{5}{35}\right)^{-2} \cdot (2^{-2}) = \frac{5^{-2}}{35^{-2}} \cdot 2^{-2} = \frac{35^2}{5^2} \cdot \frac{1}{2^2} = \frac{1225}{25} \cdot \frac{1}{4} = \frac{1225}{100} = 12.25 \text{ or } \frac{49}{4}$

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Lesson 8 Practice Period_______________________

1. Which of the following is <u>not</u> a radical expression that is equivalent to $\sqrt{96}$?

A. $\sqrt{2} \cdot \sqrt{48}$
B. $\sqrt{5} \cdot \sqrt{19}$
C. $\sqrt{6} \cdot \sqrt{16}$
D. $\sqrt{8} \cdot \sqrt{12}$

$\begin{array}{r} {}^{1}48 \\ \times\ 2 \\ \hline 96 \end{array}$ ✓ $\begin{array}{r} {}^{4}19 \\ \times\ 5 \\ \hline 95 \end{array}$ ✗ $\begin{array}{r} {}^{3}16 \\ \times\ 6 \\ \hline 96 \end{array}$ ✓ $\begin{array}{r} {}^{1}12 \\ \times\ 8 \\ \hline 96 \end{array}$ ✓

2. Which of the expressions from #1 can be rewritten in the form of an integer multiplied by the radical? *Explain.*

Option C because 16 is a perfect square so $\sqrt{6} \cdot \sqrt{16} = 4\sqrt{6}$.

For questions 3-4, write two equivalent expressions for each radical expression using the product law of radicals. One of your expressions should simplify to an integer multiplied by a radical.

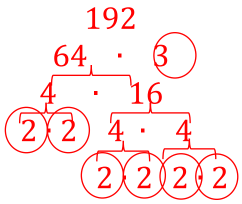
3. $\sqrt[3]{192}$ = $\sqrt[3]{64} \cdot \sqrt[3]{3} = 4\sqrt[3]{3}$

$\sqrt[3]{192}$ = $\sqrt[3]{96} \cdot \sqrt[3]{2}$

4. $\sqrt{80}$ = $\sqrt{16} \cdot \sqrt{5} = 4\sqrt{5}$

$\sqrt{80}$ = $\sqrt{2} \cdot \sqrt{40}$

5. Rewrite $\sqrt[3]{192}$ as an integer multiplied by a radical.

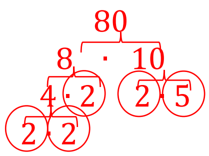


$\sqrt[3]{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3}$

$2 \cdot 2\sqrt[3]{3}$

$= 4\sqrt[3]{3}$

6. Rewrite $\sqrt{80}$ as an integer multiplied by a radical.



$\sqrt{2 \cdot 2 \cdot 2 \cdot 2 \cdot 5}$

$2 \cdot 2\sqrt{5}$

$= 4\sqrt{5}$

7. Identify all expressions that are equivalent to $2\sqrt[3]{54}$.

- $6\sqrt[3]{2}$
- $23\sqrt[3]{2}$
- $6\sqrt[3]{6}$
- $2 \cdot \sqrt[3]{2} \cdot \sqrt[3]{27}$
- $2 \cdot \sqrt[3]{6} \cdot \sqrt[3]{9}$

54

3 18

3 · 6

2 · 3

$2 \cdot \sqrt[3]{2 \cdot 3 \cdot 3 \cdot 3}$

$2 \cdot 3 \cdot \sqrt[3]{2}$

$= 6\sqrt[3]{2}$

Factors of 54: $1 \cdot 54$
$2 \cdot 27$
$3 \cdot 18$
$6 \cdot 9$

For questions 8-10, write an equivalent expression in the form of a constant multiplied by a radical.

8. $2\sqrt{200}$

$2\sqrt{100 \cdot 2}$

$10 \cdot 2\sqrt{2}$

$= 20\sqrt{2}$

9. $\sqrt[3]{40}$

40

4 · 10

2 2 2 5

$\sqrt[3]{2 \cdot 2 \cdot 2 \cdot 5}$

$= 2\sqrt[3]{5}$

10. $8\sqrt[3]{(-500)}$

$8\sqrt[3]{(-125) \cdot 4}$

$-5 \cdot 8\sqrt[3]{4}$

$= -40\sqrt[3]{4}$

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Lesson 9 Practice | Period_____________________

1. Which of the following expressions can be combined?

A. $2\sqrt{80} - 6\sqrt{20}$ $\qquad$ $2\sqrt{16 \cdot 5} - 6\sqrt{4 \cdot 5}$
B. $3\sqrt{6} + 2\sqrt{12}$ $\qquad$ $2 \cdot 4\sqrt{5} - 6 \cdot 2\sqrt{5}$
C. $5\sqrt{18} - 18\sqrt{5}$ $\qquad$ $8\sqrt{5} - 12\sqrt{5}$
D. $-4\sqrt{50} + 5\sqrt{40}$ $\qquad$ $-4\sqrt{5}$

Three students simplified the expression $2\sqrt{90} - 7\sqrt{40}$ below. Their work is shown below.

Amy	**Linda**	**Jaxson**
$2\sqrt{90} - 7\sqrt{40}$ $2\sqrt{(9 \cdot 10)} - 7\sqrt{(4 \cdot 10)}$ $6\sqrt{10} - 14\sqrt{10}$ $-8\sqrt{10}$	$2\sqrt{90} - 7\sqrt{40}$ $-5\sqrt{50}$	$2\sqrt{90} - 7\sqrt{40}$ $2\sqrt{(9 \cdot 10)} - 7\sqrt{(4 \cdot 10)}$ $18\sqrt{10} - 28\sqrt{10}$ $-10\sqrt{10}$

2. Which student simplified the expression correctly?
Amy is correct.

3. Write a note to one of the students who made an error explaining their mistake.
Answers will vary.
Linda, good try, but the expressions inside the radical cannot be subtracted.
You would first need to simplify both radicals and combine if they are like.

4. Write a note to the other student who made an error explaining their mistake.
Answers will vary.
Jaxson, you started this correctly by identifying perfect squares under the radical sign – but you forgot to take the principal square root when you brought them out.
So, you should have multiplied 2 by 3 (and not 9) and 7 by 2 (and not 4).

For question 5-10, find the sum or difference of the expression. If you cannot add or subtract the expression, explain why.

5. $\sqrt{18} - 4\sqrt{2}$

$\sqrt{9 \cdot 2} - 4\sqrt{2}$

$3\sqrt{2} - 4\sqrt{2}$

$\boxed{-\sqrt{2}}$

6. $\sqrt[3]{128} - \sqrt[3]{54}$

$\sqrt[3]{64 \cdot 2} - \sqrt[3]{27 \cdot 2}$

$4\sqrt[3]{2} - 3\sqrt[3]{2}$

$\boxed{\sqrt[3]{2}}$

7. $7\sqrt{47} + 8\sqrt{147}$

$7\sqrt{47}+56\sqrt{3}$

47 is prime and therefore must stay under the radical sign. $147 = 49 \cdot 3$, so $8\sqrt{147} = 8\sqrt{49 \cdot 3} = 8 \cdot 7\sqrt{3}$.

$\boxed{7\sqrt{47} \text{ and } 56\sqrt{3} \text{ are not like and cannot be combined (not like terms).}}$

8. $125^{\frac{1}{2}} + 320^{\frac{1}{3}}$

$\sqrt{125} + \sqrt[3]{320}$

$\sqrt{25 \cdot 5} + \sqrt[3]{64 \cdot 5}$

$5\sqrt{5} + 4\sqrt[3]{5}$

$\boxed{\text{These cannot be added since the indices are different (not like terms).}}$

9. $\sqrt{200} - 2\sqrt{48} + 4\sqrt{98}$

$\sqrt{100 \cdot 2} - 2\sqrt{16 \cdot 3} + 4\sqrt{49 \cdot 2}$

$10\sqrt{2} - 2 \cdot 4\sqrt{3} + 4 \cdot 7\sqrt{2}$

$10\sqrt{2} + 28\sqrt{2} - 8\sqrt{3}$

$\boxed{38\sqrt{2} - 8\sqrt{3}}$

10. $192^{\frac{1}{3}} + 2\left(81^{\frac{1}{3}}\right)$

$\sqrt[3]{192} + 2\sqrt[3]{81}$

$\sqrt[3]{64 \cdot 3} + 2\sqrt[3]{27 \cdot 3}$

$4\sqrt[3]{3} + 2 \cdot 3\sqrt[3]{3}$

$\boxed{10\sqrt[3]{3}}$

$4\sqrt[3]{3} + 6\sqrt[3]{3}$

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1. To multiply two ___radicals___ with the same index, multiply the integers together and then multiply the radicands together. Then simplify the radical expression.

For questions 2-7, simplify each radical expression by writing an equivalent radical expression.

2. $\sqrt{11} \cdot \sqrt{11}$

$\sqrt{121} = 11$

3. $\frac{3\sqrt{3}}{-\sqrt{48}}$

$\frac{3\sqrt{3}}{-\sqrt{16 \cdot 3}} \rightarrow \frac{3\cancel{\sqrt{3}}}{-4\cancel{\sqrt{3}}} = -\frac{3}{4}$

4. $\sqrt[3]{8} \cdot \sqrt[3]{16}$

$2 \cdot \sqrt[3]{8 \cdot 2}$
$2 \cdot 2\sqrt[3]{2} = 4\sqrt[3]{2}$

5. $\sqrt{10} \cdot \sqrt{6}$

$\sqrt{60}$
$\sqrt{4 \cdot 15} = 2\sqrt{15}$

6. $\frac{\sqrt[3]{432}}{\sqrt[3]{128}} \begin{matrix} \div 16 \\ \div 16 \end{matrix}$ $\frac{\sqrt[3]{27}}{\sqrt[3]{8}} = \frac{3}{2}$ or 1.5

7. $\sqrt[3]{98} \cdot \sqrt[3]{21}$

$$\begin{array}{ccc} 98 & & 21 \\ 2 \quad 49 & & 3 \quad 7 \\ 7 \quad 7 & & \end{array}$$

$\sqrt[3]{2 \cdot 3 \cdot 7 \cdot 7 \cdot 7}$ $= 7\sqrt[3]{6}$

8. Identify all equivalent expressions to $\sqrt{7} \cdot \sqrt{14}$.

- $\sqrt{98}$
- $7\sqrt{7}$
- $(7 \cdot 14)^{\frac{1}{2}}$
- $7\sqrt{2}$
- $14\sqrt{2}$
- $7^{\frac{1}{2}} \cdot 14^{\frac{1}{2}}$

$\sqrt{7 \cdot 14}$ or $(7 \cdot 14)^{\frac{1}{2}}$

$= \sqrt{98}$ $\quad 7^{\frac{1}{2}} \cdot 14^{\frac{1}{2}}$

$= \sqrt{2 \cdot 49}$

$= 7\sqrt{2}$

For questions 9-10, write an equivalent expression by rationalizing the denominator.

9. $\frac{-5\sqrt{3}}{2\sqrt{5}} \frac{\times\sqrt{5}}{\times\sqrt{5}} \rightarrow \frac{-5\sqrt{15}}{2\cdot\sqrt{5}\cdot\sqrt{5}} \rightarrow \frac{-5\sqrt{15}}{2\cdot 5} \rightarrow \frac{-5\sqrt{15}}{10} = \frac{-\sqrt{15}}{2}$

10. $\frac{\sqrt{27}}{3\sqrt{6}} \frac{\times\sqrt{6}}{\times\sqrt{6}} \rightarrow \frac{\sqrt{162}}{3\cdot 6} \rightarrow \frac{\sqrt{81\cdot 2}}{18} \rightarrow \frac{9\sqrt{2}}{18} \rightarrow \frac{9\sqrt{2}}{18} \frac{\div 9}{\div 9}$

$= \frac{\sqrt{2}}{2}$